

Core Invariant:  
I want to switch between internets, I will have to give a command.

Implementation Invariant:  
Some intelligence has to be there which issues the command  
eg. Location Intelligence (which house I am in, Delhi or Mumbai)  
eg. Default Last Used Workspace Identity

Maybe the trigger can be application layer. We just need to have space for it to occur.

Very One Device One Browser Session Scenario

Browser Tab 1 → Workspace 1  
Browser Tab 2 → New URL → I can pre-select a workspace, if I don't my default triggers

What are persistent workspaces?

- If I am triggering the same workspace each browser tab, who can store which url is associated with which workspace?-
- Secondly, if a url is present in only one workspace, should it by default open that workspace?
- Thirdly, can applications preserve 1 url per workspace, not allowing duplication? They will ofc get to know there is a workspace identifier, ~~by current thought.~~
- ~~Though that should have a opportunity to be hidden, like VPNs. Privacy is such a big issue. There can be a broker, but the broker will also sent something different each time.~~

Where is the Store of Workspace Data?  
It's composed of

- Identifiers
- and State

+ other things

Maybe many kinds of Identifier Storage Layers could be in the architecture.

Identifier Storage Layers by ChatGPT

| Layer                      | Can store Workspace Data? | Comments                                                                                        |
|----------------------------|---------------------------|-------------------------------------------------------------------------------------------------|
| Application Server         | ✓                         | Website stores multiple states. Requires application changes.                                   |
| Reverse Proxy / CDN        | ⚠                         | Could route requests by workspace identifier but isn't the natural owner of persistent state.   |
| ISP                        | ✗                         | Wrong layer. ISPs forward packets and generally don't maintain application-specific user state. |
| DNS                        | ✗                         | Name resolution only.                                                                           |
| Browser                    | ✓                         | Strong candidate. Already owns cookies, storage, permissions, history.                          |
| Browser Profile            | ✓                         | Existing approximation of workspaces today.                                                     |
| Browser Extension          | ✓                         | Can maintain workspace metadata, though with browser API limitations.                           |
| Operating System           | ✓                         | Could expose workspaces to all browsers and applications.                                       |
| Local Agent / Daemon       | ✓                         | Independent process managing workspace state for the device.                                    |
| Home NAS / Personal Server | ✓                         | User-controlled synchronization backend.                                                        |
| Cloud Sync Service         | ✓                         | Synchronizes workspace state across devices.                                                    |
| Identity Provider          | ⚠                         | Could associate workspaces with a user identity, but doesn't naturally own browsing state.      |
| New Internet Protocol      | ✓                         | Could define how workspace identity is communicated, while leaving storage elsewhere.           |